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3PoinTr: From Human Videos to Robot Policies with 3D Point-Track Plans

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Project page: https://adamhung60.github.io/3PoinTr/

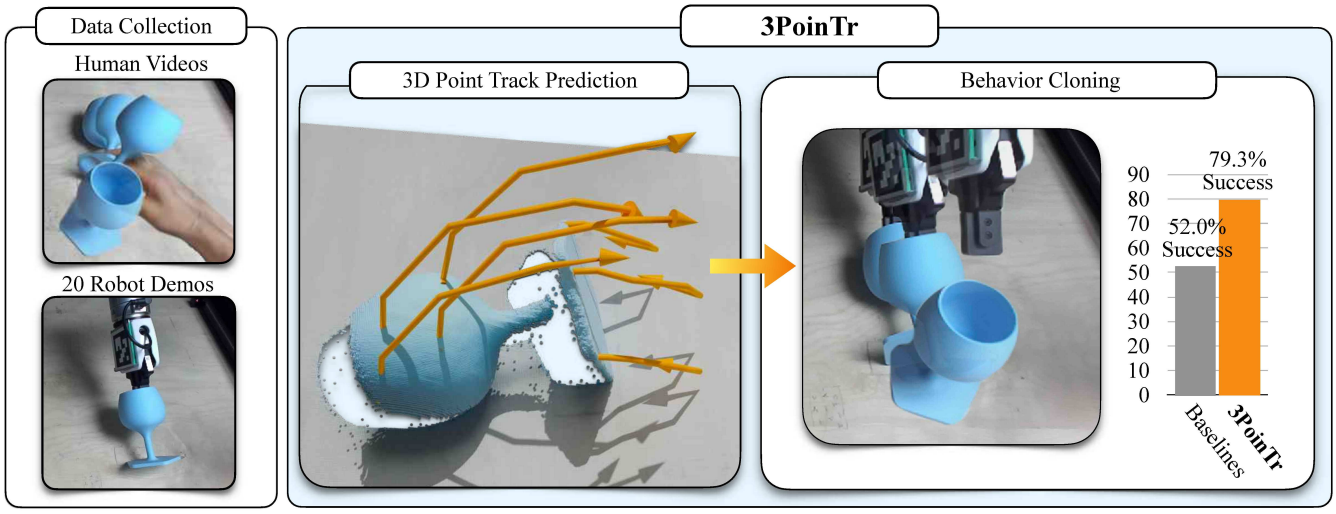

Fig. 1. We propose **3PoinTr**, a method that learns from human videos by treating 3D point tracks as global plans for robot policies. From an initial observation, 3PoinTr predicts full-task point tracks that provide a global 3D plan to a closed-loop policy. A Perceiver-IO encoder compresses this plan into a few plan tokens, which guide the policy throughout execution. We find this approach transfers exceptionally well from human demonstrations to robot execution. Across eight simulation and real-world tasks with 20 robot demonstrations per task, 3PoinTr achieves an average success rate of 79.3%, compared with 52.0% for the strongest behavior-cloning or learning-from-video baseline.

***Abstract*— Human videos could reduce the reliance on expensive robot demonstrations, but extracting task-relevant features remains challenging. We introduce 3PoinTr, a method that provides robot policies with a global 3D point-track plan predicted from human videos. Given an initial point cloud, 3PoinTr predicts dense 3D point tracks spanning the full task and compresses them into a small set of tokens that remain fixed throughout execution. At each policy query, the policy combines these plan tokens with the current observation to predict robot actions. This formulation enables the policy to extract useful motion cues from human videos. We find this formulation transfers exceptionally well, even when human demonstration motions are infeasible for robots. In controlled simulation and real-world evaluations, 3PoinTr achieves an average success rate 25.0 percentage points higher than the strongest behavior-cloning and video-pretraining baselines on real-world tasks, and an improvement of 29.6 percentage points in simulation. Ablations establish the benefits of our method, including predicting non-embodiment 3D tracks once as a global plan, Perceiver-IO compression, and a residual cross-attention mechanism. Our point-track prediction network also outperforms a strong 3D-flow baseline by retaining supervision for temporarily occluded points.**

I. Introduction

A typical approach to teaching a robot a new task is to collect teleoperation demonstrations and train a policy via behavior cloning. However, collecting sufficient demonstration data for each task is expensive [11], [12]. Human demonstration videos are much more plentiful and easy to collect, and encode the precise scene motion that defines tasks. However, humans naturally perform tasks differently from robots, often employing different strategies, trajectories, and grasps. Therefore, a key challenge is to transfer information from human demonstrations to robot control without requiring human and robot motion to precisely match.

Point tracks are a promising intermediate representation between human and robot motion because they describe task-relevant motion explicitly and can be extracted autonomously from videos. Recent methods learn point tracks from human videos and use them either to directly solve for robot actions or to condition learned policies. However, many such methods make limiting assumptions about how the predicted motion should translate into robot actions. Additionally, we find that closed-loop performance is sensitive to several critical design choices: whether tracks are predicted once as a global plan or re-predicted at each policy query, whether they are represented in 2D or 3D, and how they are compressed and integrated into the policy architecture. Table I compares prior methods along related axes.

TABLE I

POINT-TRACK METHODS FOR LEARNING ROBOT CONTROL FROM HUMAN VIDEOS. ✓/× INDICATE PRESENCE/ABSENCE OF EACH PROPERTY.

Method	3D tracks	Scene-dense tracks	Excludes embodiment points	No object segmentation	Learned policy	Tracks predicted once
Using embodiment tracks as robot actions						
Point Policy [1]	✓	×	×	✓	✓	×
Motion Track Policy [2]	×	×	×	×	✓	×
Geometrically deriving robot actions from scene motion						
Track2Act [3]	×	✓	×	✓	✓	✓
General Flow [4]	✓	×	✓	×	×	×
3DFlowAction [5]	✓	×	✓	×	×	×
TraceGen [6]	✓	✓	×	✓	×	✓
Conditioning a learned robot policy on predicted motion						
ATM [7]	×	×	×	✓	✓	×
AMPLIFY [8]	×	✓	×	✓	✓	×
μ_0 [9]	✓	×	×	×	✓	×
Im2Flow2Act [10]	×	×	✓	×	✓	✓
3PoinTr (ours)	✓	✓	✓	✓	✓	✓

Scene-dense tracks cover the scene rather than only an object or selected semantic points. *No object segmentation* means no object masks or bounding boxes are required during data processing or execution. *Learned policy* means a learned model predicts actions, rather than converting scene motion to actions geometrically. *Tracks predicted once* means one initial prediction is retained during execution, as opposed to re-predicting tracks at each time step.

We build on this line of work by treating predicted scene motion as a *global point-track plan* rather than an action target. In **3PoinTr**, tracks learned from human videos provide task-critical motion cues, while teleoperation data teaches the closed-loop policy to predict robot-specific actions that realize those cues. 3PoinTr first consumes the initial point cloud and predicts dense non-embodiment 3D point tracks that span the full task execution. A Perceiver-IO encoder [13] compresses these tracks into a small set of plan tokens, which are cached and held fixed throughout execution. At each policy query, the closed-loop policy conditions on the plan tokens together with the latest point cloud and robot state to predict the next action.

This paper contributes:

- A method for transferring motion cues from human videos to robot control without requiring human and robot trajectories to match.
- Ablations isolating the design choices that differentiate 3PoinTr from prior work: predicting point tracks once as a global plan, 3D tracks rather than 2D, predicting non-embodiment point tracks, Perceiver-IO compression into compact plan tokens, and residual track-token cross-attention in the action head.
- A 3D point-track predictor that outperforms a strong 3D-flow baseline [4] on real-world videos by retaining supervision for temporarily occluded points.

II. RELATED WORK

3PoinTr builds on prior work on behavior cloning and learning from videos.

a) Behavior cloning: Behavior cloning succeeds across a wide variety of real-world tasks [14], [15], but requires substantial teleoperated data to achieve robustness. Structured representations such as point clouds [16], [17] or keypoints [18] can improve sample efficiency, but the policy must still learn to map observations to actions from limited robot demonstrations.

b) Learning from videos: Actionless videos (videos without robot action labels) are easy to collect and encode useful task structure. Prior work uses them for visual pretraining [19], reward learning [20], affordance extraction [21], [22], and conditioning robot policies on predicted or latent video dynamics [23], [24]. Similarly, 3PoinTr transfers an explicit motion representation from human videos to robot control.

c) Learning from videos with point tracks: Point tracks provide a natural bridge between actionless videos and robot policies, encoding scene motion in an interpretable, spatially grounded form. Table I summarizes the key properties of existing point-track-based methods. We group these methods into three broad categories according to how they use tracks for robot control: using embodiment tracks as robot actions, geometrically deriving robot actions from scene motion, and conditioning a learned robot policy. Related 3D and 2D point methods fall outside this scope because they either use tracks only as auxiliary representations or do not learn robot control from human videos [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

Using embodiment tracks as robot actions. These methods represent actions as tracks of human or robot end-effector keypoints and convert predicted tracks directly into robot motion. Point Policy [1] retargets human points to robot points with a fixed geometric mapping, then trains a policy to predict these retargeted points. While this enables learning from only human demonstrations, it assumes that replaying human motion on the robot is both feasible and sufficient to solve the task. Motion Track Policy [2] instead co-trains on human and robot demonstrations using 2D manipulator-

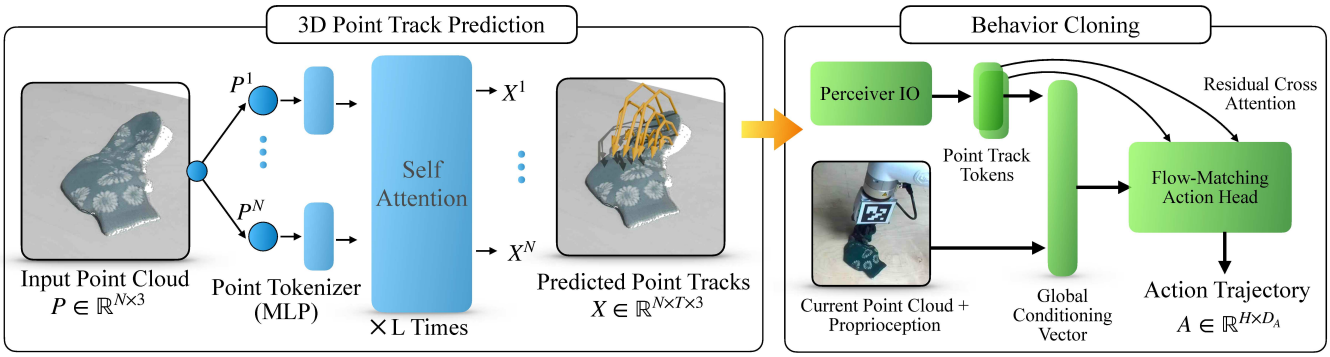

Fig. 2. Diagram of the 3PoinTr network architecture. Given an initial point cloud, a transformer predicts dense 3D point tracks spanning the full task execution. A Perceiver-IO point-track encoder compresses these tracks into a small set of learned plan tokens, which remain fixed throughout the episode. At each action step, the policy conditions on the plan tokens together with the current point cloud and robot state, enabling closed-loop action prediction. Beyond global feature conditioning, residual track-token cross-attention in every U-Net block gives the action head a direct path to the predicted object motion, encouraging a simpler mapping from point tracks to robot actions.

keypoint tracks as a shared action representation, which also assumes that humans and robots perform similar motions in similar states.

Geometrically deriving robot actions from scene motion. These methods convert predicted scene motion into robot actions through explicit geometric procedures. Doing so requires a separately specified grasp or contact strategy, supplied through heuristics, manual initialization, scripted behaviors, or external grasp-prediction modules [3], [4], [39], [5], [6]. The robot then executes the motion implied by the tracks from this selected contact configuration. Consequently, these methods cannot learn how to make contact with scenes, only how to follow a predicted motion under fixed contact.

Conditioning a learned robot policy on predicted motion. Rather than directly converting tracks into actions, these methods condition a robot policy on predicted motion, allowing it to learn robot-specific behavior from robot data. ATM [7] conditions policies on point tracks, whereas AMPLIFY [8] conditions policies on compressed motion tokens derived from point tracks. μ_0 [9] similarly conditions policies on intermediate features from a 3D point track predictor, but relies on a VLM and semantically selected keypoints.

ATM, AMPLIFY, and μ_0 all re-predict scene motion from the current observation at each action prediction step. This induces a distribution gap because intermediate robot states during execution can fall outside the human-video training distribution. Additionally, any temporarily occluded points disappear from the conditioning. In contrast, 3PoinTr predicts full-task tracks once from the initial scene and retains their compressed representation as a source of guidance throughout closed-loop execution. In our ablations (Table V), we find this choice to be crucial for real-world performance.

Im2Flow2Act [10] is another related method: it predicts object flow once and conditions a closed-loop inverse dynamics model to follow the predicted flow. However, Im2Flow2Act operates in 2D, cannot predict scene-dense point tracks, and trains its policy entirely from simulated robot play data generated with predefined action primitives. Moreover, Im2Flow2Act treats the predicted object flow as the execution target, which assumes that object motion from human videos matches how the robot should move the object.

In summary, while recent work has established point tracks as a powerful intermediate representation for learning from videos, 3PoinTr introduces a policy-conditioning recipe that does not assume human and robot motion will match, does not require object segmentations or predefined grasps, and outperforms strong video-pretrained baselines in our experiments.

III. Method

3PoinTr (Figure 2) consists of two stages: (1) learning a model that predicts dense future 3D point trajectories from a single scene observation, and (2) learning a closed-loop policy conditioned on these predicted point tracks.

A. Problem Formulation

a) Stage 1: 3D point-track prediction.: First, given an initial point cloud $P_0 \in \mathbb{R}^{N\times 3}$ of N visible scene points expressed in the camera coordinate frame with all embodiment points removed, the goal is to predict 3D point tracks: the future 3D positions of each initial point over a fixed horizon of T timesteps. Formally, we learn $f_\theta : P_0 \mapsto X$, where $X = \{X_t^i \mid i \in [0, N), t \in [0, T)\} \in \mathbb{R}^{N\times T\times 3}$, with each $X^i \in \mathbb{R}^{T\times 3}$ corresponding to a prediction for the i-th point in P_0 over T time steps.

b) Stage 2: Track-conditioned policy learning.: Given the predicted 3D point tracks X together with a current observation $o_t = (P_t, q_t)$, where $P_t \in \mathbb{R}^{N\times 3}$ is the visible point cloud at timestep t (including robot points) and q_t is the robot configuration, we train a closed-loop behavior-cloning policy that predicts a short chunk of H actions. Formally, we learn $\pi_\phi : (X, o_t) \mapsto A$, where $A \in \mathbb{R}^{H\times D_A}$ is a chunk of H actions, each with D_A dimensions. The point tracks X are produced once from the initial observation P_0 and reused throughout the episode, while o_t provides the closed-loop signal: after executing the first $H_{\text{exec}} \leq H$ predicted actions, the policy is queried again on the updated o_t.

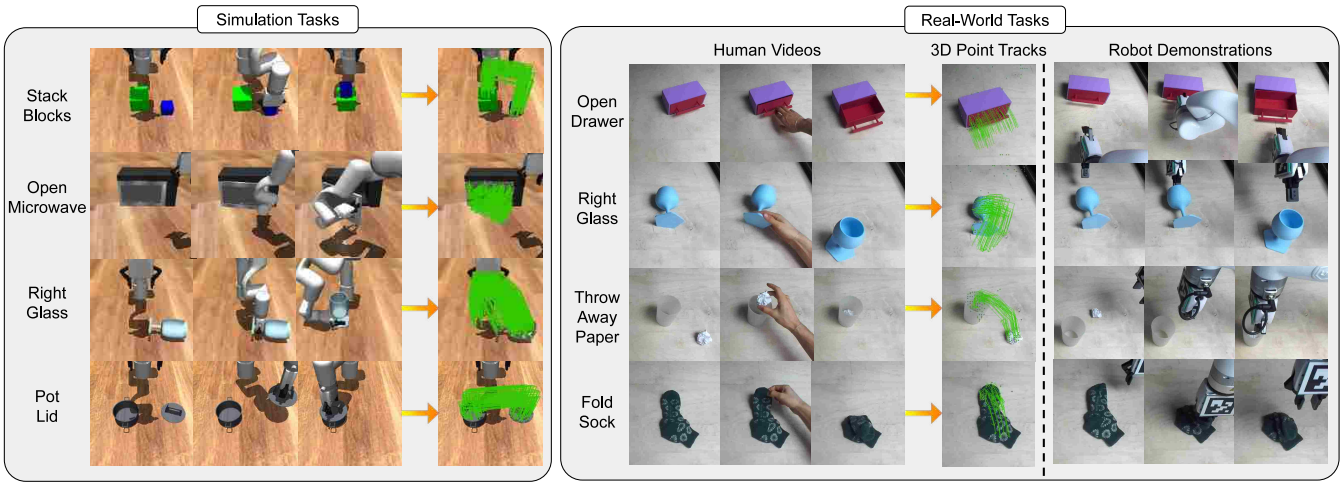

Fig. 3. Simulation and real-world tasks used for evaluation. Simulation tasks use procedurally generated robot trajectories for both video pretraining and policy learning; real-world tasks use human videos for video pretraining and robot teleoperation data for policy learning. Green tracks show 2D projections of the 3D point-track prediction targets (only tracks for moving points are visualized).

B. Network architecture and training

a) 3D point-track prediction.: We embed visible non-embodiment points with an MLP and pass the resulting tokens through a transformer with standard self-attention. A linear prediction head then outputs each point's 3D trajectory over horizon T, yielding point tracks $X \in \mathbb{R}^{N\times T\times 3}$.

During training, we subsample a fixed number of random input points from the initial point cloud, and apply per-sample unit-sphere normalization. We assume access to visibility values $m_t^i \in \{0, 1\}$, indicating whether point i is visible at timestep t. We mask losses by m_t^i, and use an L1 loss on 3D position error for all visible point-timestep pairs.

b) Track-conditioned policy learning.: During policy training, we freeze the 3D point-track predictor. Given the first frame of a task, we generate point-track predictions, and then extract compact yet informative point-track tokens with a *Perceiver-IO* architecture [13]: each predicted point trajectory X^i is first embedded into a 3D point-track token $F^i \in \mathbb{R}^{D_F}$, giving $F \in \mathbb{R}^{N\times D_F}$, and a small set of M learnable query tokens $Q \in \mathbb{R}^{M\times D_F}$ attends to F via cross-attention.

Because the policy is closed-loop, we condition it on the current point cloud P_t (including robot points) and current robot configuration q_t in addition to the persistent track tokens. We use a PointNet-style encoder following DP3 [16] to encode P_t, and a small MLP to encode q_t. The point-track, point-cloud, and configuration features are concatenated and projected into a global conditioning vector that drives a flow-matching [40] action head built on the conditional 1D U-Net architecture of Chi et al. [14].

To bias the action head to utilize the predicted object motion rather than relying on only the raw point-cloud observation, we add *residual point-track-token cross-attention* adapters throughout the U-Net in addition to the standard FiLM-style [41] global conditioning vector. At each residual block in the down path, middle blocks, and up path, the point-track tokens are linearly projected to the current U-Net channel width, and then the U-Net features query the point-track tokens. The resulting cross-attention output is added back to the U-Net features. The cross-attention output is zero-initialized, so the U-Net starts with the same function as the globally conditioned model. The action head generates action chunks composed of end-effector position, orientation, and gripper position. Actions are expressed in the camera coordinate frame. We train one policy for all simulation tasks and one policy for all real-world tasks.

IV. Experiments

We evaluate 3PoinTr in simulation and on a real robot, for both 3D point-track prediction accuracy and track-conditioned policy learning success rates.

a) Simulation Setup: In simulation, we collect demonstrations using a UFactory xArm 7 with a UFactory parallel-jaw gripper. We procedurally generate demonstrations by executing end-effector waypoints derived heuristically from randomly generated initial configurations. We evaluate on four simulation tasks (Figure 3), varying the position and orientation of objects across trials.

b) Real-World Setup: For real-world robot demonstrations and evaluation, we use an Xbox controller to teleoperate the same UFactory xArm 7 and parallel-jaw gripper as in simulation. For human video demonstration collection, a human performs the manipulation tasks. We calibrate a ZED Mini camera to the scene using eye-to-hand calibration, and keep the camera static.

We evaluate on four real-world tasks (Figure 3), varying the position and orientation of objects across trials. Although human and robot demonstrations share similar initial and goal states, the robot gripper and human hand do not need to have the same configurations or trajectories throughout the task. Instead of trying to mimic the intended downstream robot motion, the human demonstrator performs the tasks in a manner they see fit. For example, for the *Right Glass* task, the human grasps the glass by the stem, which leads

TABLE II

POINT-TRACK AVERAGE DISPLACEMENT ERROR (ADE) AND ADE OF THE 5 % OF POINTS THAT MOVE THE MOST (5% ADE) FOR 3POINTR AND GENERAL FLOW [4]; LOWER IS BETTER AND UNITS ARE MILLIMETERS. ON REAL-WORLD TASKS, 3POINTR LOWERS AVERAGE ADE BY **28 %** AND 5% ADE BY **44 %**, OUTPERFORMING GENERAL FLOW ON EVERY TASK AND METRIC BY RETAINING SUPERVISION FOR TEMPORARILY OCCLUDED POINTS. IN SIMULATION, WHERE THERE IS NO OCCLUSION, THE METHODS ARE COMPETITIVE.

Simulation Tasks					**Real-World Tasks**				
Task	**General Flow**		**3PoinTr**		**Task**	**General Flow**		**3PoinTr**	
	ADE↓	5% ADE↓	ADE↓	5% ADE↓		ADE↓	5% ADE↓	ADE↓	5% ADE↓
Block Stack	**0.47**	**2.56**	0.85	5.98	Open Drawer	2.54	16.37	**2.18**	**13.19**
Open Microwave	**0.68**	**6.04**	1.79	6.79	Right Glass	3.34	30.38	**2.29**	**18.55**
Right Glass	1.96	28.82	**1.66**	**15.86**	Throw Away Paper	2.40	22.87	**1.47**	**8.17**
Pot Lid	1.18	6.89	**0.77**	**4.83**	Fold Sock	1.56	10.09	**1.14**	**4.68**
Average	**1.07**	11.08	1.27	**8.37**	**Average**	2.46	19.93	**1.77**	**11.15**

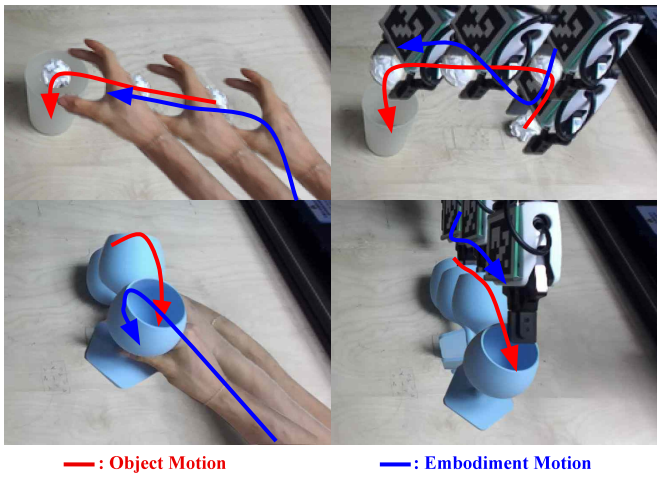

Fig. 4. Example human vs. robot trajectories: 3PoinTr learns scene motion from human videos (left) and derives a global point-track plan, while robot demonstrations teach the closed-loop policy how to realize that plan through embodiment-specific actions (right). This separation lets the human and robot use different grasps, trajectories, and even strategies.

to singular configurations if the robot attempts to mimic; instead, the robot inserts its gripper into the glass and lifts the rim such that the glass pivots about its base. Although the general strategies for the other tasks are more similar between human and robot, the angles of approach, grasp points, and trajectories differ between the human and robot embodiments. Figure 4 shows examples of these embodiment-induced trajectory differences.

A. Implementation Details

a) Data collection: For both simulation and real-world domains, we train on a mix of action-labeled robot trajectories and actionless videos. In simulation, we collect actionless videos by removing action labels from robot demonstrations, whereas in real-world experiments, we collect human demonstration videos.

Our data collection process extracts 3D point tracks for a dense grid of points initialized at the first image frame of each demonstration. The real-world data additionally includes per-frame visibility masks. In simulation, we obtain tracks from ground-truth rigid-body poses. For real-world videos, we estimate 2D tracks and visibility with CoTracker3 [42], estimate stereo depth with Foundation-Stereo [43], and lift tracks to 3D. We segment human-arm points with SAM3 [44] and mark them invisible. We resample all trajectories uniformly in normalized time. We use coarse trajectories of $\sim$10–16 timesteps as the training targets for the 3D point-track predictor, spanning the complete task execution. We use finer trajectories of $\sim$30–50 timesteps as the action targets for the closed-loop policy. In real-world experiments, we train point-track prediction networks solely on human videos, as we find that this provides consistent conditioning for the policies.

b) Point-track prediction network: We use a 3-layer MLP with output dimension 256 and GELU activations to tokenize input points. We use two decoder blocks with 4 attention heads with head dimension 64. In between the blocks we use a feed-forward network with expansion ratio 4 and GELU activation. After the decoder, a linear head predicts per-point 3D trajectories over the full task-execution horizon. At both training and test times, we randomly subsample 2048 points from the initial point cloud. We add Gaussian noise with $\sigma = 0.01$ to input point clouds as data augmentation.

c) Track-conditioned policy network: The 2048 per-point trajectory tokens are projected to 128-D, and 4 learned Perceiver-IO queries cross-attend to them to produce 4 point-track tokens. These point-track tokens are then used for both global conditioning and residual cross-attention in the U-Net. The global conditioning vector is formed by concatenating three features: (i) the flattened point-track tokens, (ii) a 64-D PointNet feature of the current point cloud, and (iii) a 32-D feature of the current robot configuration, followed by a projection to 128-D.

The flow-matching action head predicts an action chunk $A \in \mathbb{R}^{H \times D_A}$ with $D_A = 10$, where H is the prediction horizon. Each action contains a 3D end-effector position, a 6D continuous orientation representation [45], and a 1D gripper command. Point clouds and positional actions are jointly centered by the initial point cloud mean and scaled by its 90th-percentile point radius; rotations are left unchanged and gripper values use a fixed scale. For multitask learning, we sample tasks uniformly and, after one warmup epoch, inversely weight each task's loss by its mean warmup loss.

TABLE III

SIMULATION SUCCESS RATES (%) OVER 200 ROLLOUTS PER TASK. POLICIES USE 20, 50, OR 100 ACTION-LABELED DEMONSTRATIONS; VIDEO-PRETRAINED METHODS ADDITIONALLY USE 100 ACTIONLESS VIDEOS PER TASK. WITH 20 DEMONSTRATIONS, 3POINTR ACHIEVES THE HIGHEST SUCCESS RATE ON EVERY TASK AND IMPROVES AVERAGE SUCCESS BY **29.6** PERCENTAGE POINTS OVER THE STRONGEST BASELINE; IT ALSO ACHIEVES THE HIGHEST AVERAGE SUCCESS RATE AT 50 AND 100 DEMONSTRATIONS.

Task	**AMPLIFY** [8]			**DP** [14]			**ATM** [7]			**DP3** [16]			**3PoinTr**		
# Demonstrations	20	50	100	20	50	100	20	50	100	20	50	100	20	50	100
Block Stack	0.5	3.5	2.0	17.0	30.5	60.5	11.0	47.5	67.0	25.5	41.5	57.0	**66.0**	**84.5**	**94.0**
Right Glass	2.0	6.0	9.0	16.5	27.5	76.5	8.0	25.0	32.5	21.0	43.0	70.0	**51.0**	**77.0**	**97.0**
Pot Lid	3.0	4.0	7.0	42.5	51.0	58.0	27.5	51.0	54.5	51.5	60.0	77.5	**67.5**	**91.0**	**93.0**
Open Microwave	7.0	2.5	8.0	17.5	35.0	81.5	70.0	**93.5**	**99.0**	43.0	48.0	56.0	**75.0**	78.0	89.0
Average	3.1	4.0	6.5	23.4	36.0	69.1	29.1	54.3	63.3	35.3	48.1	65.1	**64.9**	**82.6**	**93.3**

TABLE IV

REAL-WORLD SUCCESS RATES OVER 20 ROLLOUTS PER TASK, USING 20 ROBOT DEMONSTRATIONS AND 50 HUMAN VIDEOS PER TASK. 3POINTR SUCCEEDS ON 18.8/20 ROLLOUTS ON AVERAGE, OUTPERFORMING THE STRONGEST BASELINE, DP3, BY **25.0** PERCENTAGE POINTS AND ACHIEVING THE HIGHEST SUCCESS RATE ON EVERY TASK.

Method	**Open Drawer**	**Right Glass**	**Throw Away Paper**	**Fold Sock**	**Average**
ATM	6/20	3/20	0/20	7/20	4.0/20
DP3	14/20	18/20	9/20	14/20	13.8/20
3PoinTr	**20/20**	**20/20**	**18/20**	**17/20**	**18.8/20**

B. Baselines

a) 3D Point-Track Prediction Baseline: We compare to a strong 3D flow prediction method, General Flow [4], by retraining the General Flow network from scratch on our dataset.

b) Policy Learning Baselines: We compare to two behavior cloning baselines that do not use any video pretraining, and two motion-prediction-based methods pretrained on actionless videos:

- *Diffusion Policy (DP)* [14]: behavior cloning with a diffusion action head conditioned on RGB images and robot poses.
- *3D Diffusion Policy (DP3)* [16]: Diffusion Policy conditioned on 3D point clouds instead of images.
- *ATM* [7]: predicts future 2D point tracks and uses them for policy learning. Following ATM, we train its track predictor on both robot and human videos for real-world tasks.
- *AMPLIFY* [8]: learns video dynamics in a compressed latent space, then learns inverse dynamics from limited action-labeled demonstrations.

We do not benchmark methods that use embodiment tracks as robot actions or geometrically convert scene motion into actions in Table I because they require embodiment correspondences or predefined contacts; they also directly execute predicted motion, which can be infeasible or collision-prone when human and robot motion differ. We do not benchmark Im2Flow2Act [10] because its policy requires simulated play data collected in constructed environments with manually defined exploration primitives, whereas our setting focuses on learning solely from actionless videos and robot demonstrations.

We evaluate all baselines in simulation. We select the best-performing baseline from each class (DP3 for behavior cloning, and ATM for flow-based video pretraining) for evaluation on real-world tasks.

C. Results: 3D Point-Track Prediction

We evaluate 3PoinTr's 3D point-track predictions on all simulation and real-world tasks using 3D Average Displacement Error (ADE [4]) in millimeters. We report total ADE over all points and ADE over the 5 % of points that move the most (5 % ADE). We propose the 5 % ADE metric to reflect task-critical motions, as total ADE is dominated by static background points. Results are computed over 100 unseen simulation samples per task and 20 unseen real-world samples per task. Although each method predicts full trajectories for every input point, errors are computed only for visible ground-truth point-timestep pairs (i, t).

Table II shows 3PoinTr outperforming General Flow in both metrics on every real-world task, with average error reductions of **28 %** and **44 %**, respectively. The primary advantage of 3PoinTr is that it trains on data General Flow ignores. Real-world points are often temporarily occluded; General Flow removes any trajectory with invisible point-timestep pairs during preprocessing, whereas 3PoinTr retains all trajectories and masks losses for individual invisible point-timestep pairs. This provides additional supervision over task-critical object points that are temporarily occluded during manipulation. For example, in the *Throw Away Paper* task, every paper point is invisible by the final frame, so General Flow receives no supervision over paper motion. As a result, 3PoinTr generates more informative point tracks from our real-world manipulation videos. In simulation, where all points are visible and supervision is identical, 3PoinTr's single-transformer decoder remains competitive with General Flow's more complex PointNeXt-conditioned [46] conditional VAE.

D. Results: Sample-Efficient Policy Learning

Next, we evaluate 3PoinTr's 3D-point-track-conditioned policy performance. We display success rates across simulated and real tasks in Tables III and IV. We evaluate

TABLE V

ABLATION OF 3POINTR DESIGN CHOICES DEFINED IN SECTION IV-E. SIMULATION VARIANTS (A) REPLACE 3D TRACKS WITH 2D TRACKS, REPLACE PERCEIVER-IO WITH MEAN POOLING, REMOVE RESIDUAL U-NET CROSS-ATTENTION, OR TRAIN THE TRACK PREDICTOR WITHOUT EXTRA ACTIONLESS VIDEOS. THESE ARE TRAINED ON 20 ROBOT DEMONSTRATIONS AND 100 ACTIONLESS VIDEOS PER TASK, AND EVALUATED ON 200 ROLLOUTS PER TASK. REAL-WORLD VARIANTS (B) INCLUDE EMBODIMENT POINTS IN THE TRACK TARGETS OR RE-PREDICT TRACKS AT EVERY POLICY QUERY. THESE ARE TRAINED ON 20 ROBOT DEMONSTRATIONS AND 50 HUMAN VIDEOS PER TASK, AND EVALUATED ON 20 ROLLOUTS PER TASK. 3POINTR ACHIEVES THE HIGHEST AVERAGE SUCCESS RATE IN BOTH DOMAINS, DEMONSTRATING THE CONTRIBUTION OF EACH DESIGN CHOICE.

(a) Simulation

Task	2D Tracks	No Perc.	No Xattn	No Videos	3PoinTr
Block Stack	35.0	24.5	56.0	**69.0**	66.0
Right Glass	26.5	25.0	41.0	19.5	**51.0**
Pot Lid	44.0	55.5	62.5	57.5	**67.5**
Open Microwave	37.5	29.0	70.0	60.0	**75.0**
Average	35.8	33.5	57.4	51.5	**64.9**

(b) Real world

Task	Incl. Emb. Tracks	Per-Step Tracks	3PoinTr
Open Drawer	19/20	16/20	**20/20**
Right Glass	**20/20**	13/20	**20/20**
Throw Away Paper	10/20	6/20	**18/20**
Fold Sock	12/20	13/20	**17/20**
Average	15.3/20	12.0/20	**18.8/20**

multitask policies trained on all real-world tasks with 20 robot demonstrations and 50 actionless human videos per task, and separately on all simulation tasks with 20, 50, and 100 action-labeled demonstrations and 100 actionless videos per task. Behavior cloning baselines only use action-labeled data. 3PoinTr achieves the highest success rate on every task at 20 robot demonstrations, and the highest average success rate across tasks at 50 and 100 demonstrations.

Outperforming the behavior cloning baselines (DP and DP3) suggests that 3PoinTr is able to extract useful task information from 3D point-track predictions, thus effectively leveraging additional supervision from human videos. We attribute the gains over ATM and AMPLIFY to our design choices, which we isolate in Section IV-E. Overall, these choices yield substantially stronger average performance while making fewer assumptions about visual and kinematic alignment between human and robot demonstrations.

E. Results: Ablations

We ablate key design decisions across simulated and real tasks.

a) Simulation ablations: In simulation, 3PoinTr achieves the highest average success rate across ablations, as shown in Table V.

- *No Perceiver-IO*: replaces the Perceiver-IO encoder with mean pooling. This produces the largest drop in average success among simulation ablations.
- *No U-Net Xattn*: removes residual point-track-token cross-attention from the U-Net and reduces performance across all tasks.
- *2D Point Tracks*: predicts and conditions on 2D tracks instead of 3D tracks. This substantially lowers average success, demonstrating the importance of 3D scene representations.
- *No Extra Videos*: trains the track predictor only on the 20 robot demonstrations. Its strong performance shows that point-track prediction itself is a useful inductive bias, while the full model's higher average success establishes the benefit of training on extra actionless videos.

b) Real-world ablations: We conduct two real-world ablations targeting design choices that are specifically relevant when learning from human videos alongside robot demonstrations. For both, we train the track predictor on human and robot videos to reduce train-to-test distribution shift. These variants adopt common choices from prior methods (Table I)—using demonstrator embodiment tracks [1], [2], [3], [6], [7], [8] and re-predicting tracks during execution [1], [2], [4], [39], [7], [8]—to highlight the benefits of our design.

- *Incl. Emb. Tracks*: includes demonstrator embodiment points in the track-prediction targets. Its lower performance is consistent with embodiment-specific tracks being less transferable when human and robot motion differ.
- *Per-Step Tracks*: retains embodiment masking but re-predicts tracks from the current observation at every policy query over the action horizon H, rather than predicting full-task tracks once from the initial observation. We attribute its lower performance to two factors: repeated track prediction exposing the track predictor to intermediate robot states that differ from the human demonstration distribution, and occlusions removing important object and scene point tracks from the policy conditioning.

V. CONCLUSION

We introduced 3PoinTr, an effective recipe for learning manipulation policies from human videos with a small number of teleoperated robot demonstrations. A predictor produces dense 3D tracks over non-embodiment scene points once as a global plan, a Perceiver-IO encoder compresses them into plan tokens, and residual cross-attention conditions a closed-loop policy on those tokens throughout robot execution.

Across our controlled simulation and real-world tasks, this recipe outperforms behavior-cloning and video-pretraining baselines. The 3D point-track predictor also improves over a strong flow baseline on real-world videos, and ablations isolate the contributions of each design choice. Together, the results establish 3D point-track plans for closed-loop policies as an effective approach to learning from human videos, even when human and robot motion differs.

VI. Limitations

Our experiments are conducted across a relatively small set of tasks in a controlled laboratory setting with a static camera. We therefore do not demonstrate transfer across differing tasks, viewpoints, or environments. The real-world pipeline also relies on stereo depth estimation, point tracking, and human-arm segmentation. We also omit natural language conditioning.

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